

How Ricardian Are We?

U of Oklahoma Economics Seminar, 17 April 2026

Jonathan J. Adams ¹ Christian Matthes ²

¹Federal Reserve Bank of Kansas City

²University of Notre Dame

The views expressed herein are those of the authors and do not necessarily represent the views of the Federal Reserve Bank of Kansas City or the Federal Reserve System.

Introduction

Motivation

- How Ricardian is the average household?

Motivation

- How Ricardian is the average household?
 - What share of bonds is perceived as net wealth?

Motivation

- How Ricardian is the average household?
 - What share of bonds is perceived as net wealth?
 - What share of government borrowing is offset by private savings?

Motivation

- How Ricardian is the average household?
 - What share of bonds is perceived as net wealth?
 - What share of government borrowing is offset by private savings?
 - What share of transfers is consumed?

Motivation

- How Ricardian is the average household?
 - What share of bonds is perceived as net wealth?
 - What share of government borrowing is offset by private savings?
 - What share of transfers is consumed?
- Critical for policy: fiscal stimulus, optimal taxation, debt management, crisis response, etc.

Motivation

- How Ricardian is the average household?
 - What share of bonds is perceived as net wealth?
 - What share of government borrowing is offset by private savings?
 - What share of transfers is consumed?
- Critical for policy: fiscal stimulus, optimal taxation, debt management, crisis response, etc.
- ... but we have no idea!

Motivation

- How Ricardian is the average household?
 - What share of bonds is perceived as net wealth?
 - What share of government borrowing is offset by private savings?
 - What share of transfers is consumed?
- Critical for policy: fiscal stimulus, optimal taxation, debt management, crisis response, etc.
- ... but we have no idea!
 - Many attempts in 1980s to answer by estimating the *consumption function*

Motivation

- How Ricardian is the average household?
 - What share of bonds is perceived as net wealth?
 - What share of government borrowing is offset by private savings?
 - What share of transfers is consumed?
- Critical for policy: fiscal stimulus, optimal taxation, debt management, crisis response, etc.
- ... but we have no idea!
 - Many attempts in 1980s to answer by estimating the *consumption function*
 - OLS fails; time series research ceased without resolution

Motivation

- How Ricardian is the average household?
 - What share of bonds is perceived as net wealth?
 - What share of government borrowing is offset by private savings?
 - What share of transfers is consumed?
- Critical for policy: fiscal stimulus, optimal taxation, debt management, crisis response, etc.
- ... but we have no idea!
 - Many attempts in 1980s to answer by estimating the *consumption function*
 - OLS fails; time series research ceased without resolution
 - Micro studies estimate generic MPC, but we need the *MPC out of Debt-Financed Transfers* (missing intercept problem)

Our Contributions

1. Posit that non-Ricardianism is due to failure of rational expectations $\implies$
derive a *behavioral consumption function*

Our Contributions

1. Posit that non-Ricardianism is due to failure of rational expectations $\implies$ derive a *behavioral consumption function*
 - Fairly general: derived from budget constraints + the Euler equation

Our Contributions

1. Posit that non-Ricardianism is due to failure of rational expectations $\implies$ derive a *behavioral consumption function*
 - Fairly general: derived from budget constraints + the Euler equation
 - Nests other mechanisms (e.g. liquidity constraints, OLG, taxes)

Our Contributions

1. Posit that non-Ricardianism is due to failure of rational expectations $\implies$ derive a *behavioral consumption function*
 - Fairly general: derived from budget constraints + the Euler equation
 - Nests other mechanisms (e.g. liquidity constraints, OLG, taxes)
2. OLS fails $\implies$ estimate by IV

Our Contributions

1. Posit that non-Ricardianism is due to failure of rational expectations $\implies$ derive a *behavioral consumption function*
 - Fairly general: derived from budget constraints + the Euler equation
 - Nests other mechanisms (e.g. liquidity constraints, OLG, taxes)
2. OLS fails $\implies$ estimate by IV
 - Demand shocks cause OLS bias towards Ricardian equivalence

Our Contributions

1. Posit that non-Ricardianism is due to failure of rational expectations $\implies$ derive a *behavioral consumption function*
 - Fairly general: derived from budget constraints + the Euler equation
 - Nests other mechanisms (e.g. liquidity constraints, OLG, taxes)
2. OLS fails $\implies$ estimate by IV
 - Demand shocks cause OLS bias towards Ricardian equivalence
 - Barnichon-Mesters: structural macro equations can be estimated using macro shocks as IVs!

Our Contributions

1. Posit that non-Ricardianism is due to failure of rational expectations $\implies$ derive a *behavioral consumption function*
 - Fairly general: derived from budget constraints + the Euler equation
 - Nests other mechanisms (e.g. liquidity constraints, OLG, taxes)
2. OLS fails $\implies$ estimate by IV
 - Demand shocks cause OLS bias towards Ricardian equivalence
 - Barnichon-Mesters: structural macro equations can be estimated using macro shocks as IVs!
 - Consumption function has many endogenous regressions, so we assemble a large number of well-identified shocks from the literature

Our Contributions (cont.)

3. Develop a method to estimate structural macro equations: the **B**ayesian **H**ybrid **IV** **E**stimator (**B-HIVE**)

Our Contributions (cont.)

3. Develop a method to estimate structural macro equations: the **B**ayesian **H**ybrid **IV** **E**stimator (**B-HIVE**)
 - Useful with many endogenous regressors + many IVs

Our Contributions (cont.)

3. Develop a method to estimate structural macro equations: the **Bayesian Hybrid IV Estimator (B-HIVE)**
 - Useful with many endogenous regressors + many IVs
 - State space naturally handles missing data, mixed frequencies

Our Contributions (cont.)

3. Develop a method to estimate structural macro equations: the **Bayesian Hybrid IV Estimator (B-HIVE)**
 - Useful with many endogenous regressors + many IVs
 - State space naturally handles missing data, mixed frequencies
 - Expectations appearing in structural eqns are internally consistent

Our Contributions (cont.)

3. Develop a method to estimate structural macro equations: the **Bayesian Hybrid IV Estimator (B-HIVE)**
 - Useful with many endogenous regressors + many IVs
 - State space naturally handles missing data, mixed frequencies
 - Expectations appearing in structural eqns are internally consistent
 - Usual Bayesian benefits (e.g. penalization, informed priors) especially helpful with weak instruments

Our Contributions (cont.)

3. Develop a method to estimate structural macro equations: the **Bayesian Hybrid IV Estimator (B-HIVE)**
 - Useful with many endogenous regressors + many IVs
 - State space naturally handles missing data, mixed frequencies
 - Expectations appearing in structural eqns are internally consistent
 - Usual Bayesian benefits (e.g. penalization, informed priors) especially helpful with weak instruments
4. Apply to US data: *How Ricardian Are We?*

Our Contributions (cont.)

3. Develop a method to estimate structural macro equations: the **Bayesian Hybrid IV Estimator (B-HIVE)**
 - Useful with many endogenous regressors + many IVs
 - State space naturally handles missing data, mixed frequencies
 - Expectations appearing in structural eqns are internally consistent
 - Usual Bayesian benefits (e.g. penalization, informed priors) especially helpful with weak instruments
4. Apply to US data: *How Ricardian Are We?*
 - **Not very!**

Our Contributions (cont.)

3. Develop a method to estimate structural macro equations: the **Bayesian Hybrid IV Estimator (B-HIVE)**
 - Useful with many endogenous regressors + many IVs
 - State space naturally handles missing data, mixed frequencies
 - Expectations appearing in structural eqns are internally consistent
 - Usual Bayesian benefits (e.g. penalization, informed priors) especially helpful with weak instruments
4. Apply to US data: *How Ricardian Are We?*
 - **Not very!**
 - Households only internalize 1/5 of future taxes

Our Contributions (cont.)

3. Develop a method to estimate structural macro equations: the **Bayesian Hybrid IV Estimator (B-HIVE)**
 - Useful with many endogenous regressors + many IVs
 - State space naturally handles missing data, mixed frequencies
 - Expectations appearing in structural eqns are internally consistent
 - Usual Bayesian benefits (e.g. penalization, informed priors) especially helpful with weak instruments
4. Apply to US data: *How Ricardian Are We?*
 - **Not very!**
 - Households only internalize 1/5 of future taxes
 - GE model: government borrowing crowds out capital

Theoretical Framework

“The debts of a nation are debts due from the right hand to the left.”

— David Ricardo (1817)

(paraphrasing Jean-François Melon (1734))

“The debts of a nation are debts due from the right hand to the left.”

— David Ricardo (1817)

(paraphrasing Jean-François Melon (1734))

“The people who pay the taxes never so estimate them, and therefore do not manage their private affairs accordingly.”

— David Ricardo (1820)

Behavioral Consumption Function Ingredients

- Government budget constraint:

$$B_{t-1} + G_t = T_t + Q_t B_t$$

- Household budget constraint:

$$B_{t-1} + R_t^K K_{t-1} + Y_t^N = C_t + T_t + Q_t B_t + K_t$$

- Household Euler Equation

$$Q_t = \beta \tilde{\mathbb{E}}_t \left[\frac{u'(C_{t+1})}{u'(C_t)} \right] + Z_t^d$$

Notes: $\tilde{\mathbb{E}}_t$ is a (possibly) non-rational expectation operator, Z_t^d an exogenous wedge

Behavioral Consumption Function Ingredients (Linearized)

- Government budget constraint:

$$b_{t-1} = \tau_t - g_t + \bar{B}q_t + \beta b_t$$

- Household budget constraint:

$$n_{t-1} + y_t = c_t + \tau_t + q_t\bar{B} + \beta n_t$$

- Household Euler Equation

$$q_t = \beta \tilde{\mathbb{E}}_t[\gamma(c_t - c_{t+1})] + z_t^d$$

Notes: n_t is household financial net worth, steady state returns are β^{-1} ,
normalize $\bar{C} = 1$

Behavioral Consumption Function: Expectations

- What is $\tilde{\mathbb{E}}_t$?
 - Distorts the rational expectation $\mathbb{E}_t$; relatively general
 - Can apply to all time series or subset
 - Naive or sophisticated (i.e. without or with L.I.E.)
- v_t and $\tilde{v}_t$ denote present values, rational and perceived, e.g.

$$v_t^\tau = \tau_t + \beta \mathbb{E}_t[v_{t+1}^\tau] \qquad \tilde{v}_t^\tau = \tau_t + \beta \tilde{\mathbb{E}}_t[v_{t+1}^\tau]$$

- Assumption: perceived P.V. of future taxes is proportional to true P.V.

$$\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] = \theta \mathbb{E}_t[v_{t+1}^\tau]$$

θ is the *behavioral attenuation* (key parameter)

The Behavioral Consumption Function

Proposition

If expectations satisfy $\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] = \theta \mathbb{E}_t[v_{t+1}^\tau]$ then consumption is given by

$$c_t = (1-\beta) \left(n_{t-1} - \theta b_{t-1} + \tilde{v}_t^y - (1-\theta)\tau_t - \theta v_t^g + \theta \bar{B} v_t^q \right) + \left(\frac{1}{\gamma} - (1-\beta)\bar{B} \right) \tilde{v}_t^q + \zeta_t$$

- ζ_t exogenous demand shock (determined by intertemporal wedges)
- Isomorphic to other non-Ricardian mechanisms (more on that later)

Empirical Strategy

Why Do We Need Time Series Regressions?

- Many micro studies estimate the MPC

Why Do We Need Time Series Regressions?

- Many micro studies estimate the MPC
- ... but to tell if people are Ricardian, you need the *MPC out of Debt-Financed Transfers* (MPC-DFT)

Why Do We Need Time Series Regressions?

- Many micro studies estimate the MPC
- ... but to tell if people are Ricardian, you need the *MPC out of Debt-Financed Transfers* (MPC-DFT)
- Cannot be estimated in (existing) cross-sectional studies because of the *missing intercept problem*

Why Do We Need Time Series Regressions?

- Many micro studies estimate the MPC
- ... but to tell if people are Ricardian, you need the *MPC out of Debt-Financed Transfers* (MPC-DFT)
- Cannot be estimated in (existing) cross-sectional studies because of the *missing intercept problem*
- So we will estimate the consumption function with time series data

Empirical Strategy: Instrumental Variables

- Problem: cannot estimate by OLS if demand shock ζ_t affects endogenous variables (likely)

Empirical Strategy: Instrumental Variables

- Problem: cannot estimate by OLS if demand shock ζ_t affects endogenous variables (likely)
- Solution: use macro shocks from the literature as IVs (Barnichon-Mesters)

Empirical Strategy: Instrumental Variables

- Problem: cannot estimate by OLS if demand shock ζ_t affects endogenous variables (likely)
- Solution: use macro shocks from the literature as IVs (Barnichon-Mesters)
- This plan brings several challenges: develop **B-HIVE** to resolve

Challenges

- Missing data: well-identified macro shocks have mixed coverage, missing observations

Challenges

- Missing data: well-identified macro shocks have mixed coverage, missing observations
 - Especially problematic here: many endogenous regressors

Challenges

- Missing data: well-identified macro shocks have mixed coverage, missing observations
 - Especially problematic here: many endogenous regressors
- Famously weak instruments

Challenges

- Missing data: well-identified macro shocks have mixed coverage, missing observations
 - Especially problematic here: many endogenous regressors
- Famously weak instruments
- Expectations appear in the structural equation

Challenges

- Missing data: well-identified macro shocks have mixed coverage, missing observations
 - Especially problematic here: many endogenous regressors
- Famously weak instruments
- Expectations appear in the structural equation
 - Especially problematic here: long horizons in v_t^y etc.

The Bayesian Hybrid Instrumental Variable Estimator



The Bayesian Hybrid Instrumental Variable Estimator



B-HIVE: State-Space Components

1. A structural equation:

$$c_t = \phi_0 + \phi_n n_{t-1} + \phi_b b_{t-1} + \phi_\tau \tau_t + \sum_{j \in \{y, g, q\}} \tilde{\phi}_j v_t^j + \zeta_t$$

B-HIVE: State-Space Components

1. A structural equation:

$$c_t = \phi_0 + \phi_n n_{t-1} + \phi_b b_{t-1} + \phi_\tau \tau_t + \sum_{j \in \{y, g, q\}} \tilde{\phi}_j v_t^j + \zeta_t$$

2. A VAR structure for X_t (RHS variables + other observables)

$$X_t = \mu_X + A_1 X_{t-1} + \cdots + A_p X_{t-p} + G \varepsilon_t \quad , \quad \varepsilon_t \sim N(0, I)$$

B-HIVE: State-Space Components

1. A structural equation:

$$c_t = \phi_0 + \phi_n n_{t-1} + \phi_b b_{t-1} + \phi_\tau \tau_t + \sum_{j \in \{y, g, q\}} \tilde{\phi}_j v_t^j + \zeta_t$$

2. A VAR structure for X_t (RHS variables + other observables)

$$X_t = \mu_X + A_1 X_{t-1} + \cdots + A_p X_{t-p} + G \varepsilon_t \quad , \quad \varepsilon_t \sim N(0, I)$$

3. A measurement equation relating instruments w_t to structural shocks ε_t

$$w_t = \mu_w + M_X X_{t-1} + M \varepsilon_t + \eta_t \quad \eta_t \sim N(0, \Sigma_\eta)$$

Coefficients can be time-dependent.

Application

- National accounts, everything at household level
 - y_t is personal income (less transfers), τ_t is personal taxes (net of transfers), g_t is expenditures net of residual taxes, n_t is household net worth
 - Normalize everything relative to a nominal GDP trend
- Many many shocks (next slide)
- Baseline: let $\tilde{v}_t^y, \tilde{v}_t^q$ be proportional to rational expectations
 - Later: augment with survey forecasts

Classification: External, HFI, Narrative, SVAR

Monetary Policy Shocks: Jarociński and Karadi (2020), Miranda-Agrippino and Ricco (2021), Bauer and Swanson (2023), Swanson (2024), Aruoba and Drechsel (2024), Drechsel (2024)

Government Spending Shocks: Fisher and Peters (2010), Ramey (2016), Romer and Romer (2016), Fieldhouse et al. (2018), Fieldhouse and Mertens (2023)

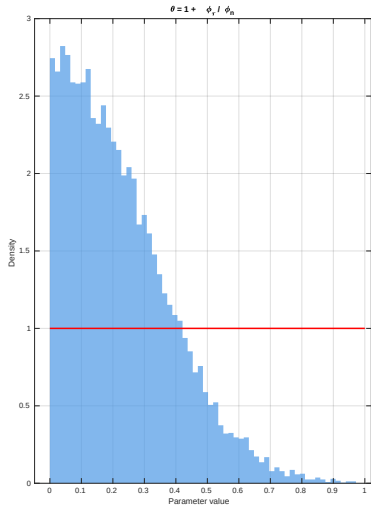
Tax/Borrowing Shocks: Leeper et al. (2012), Phillot (2025), Mertens and Ravn (2012), Lieb et al. (2024)

Technology Shocks: Fernald (2014), Miranda-Agrippino et al. (2025)

Oil Shocks: Kilian (2008), Känzig (2021), Baumeister and Hamilton (2019)

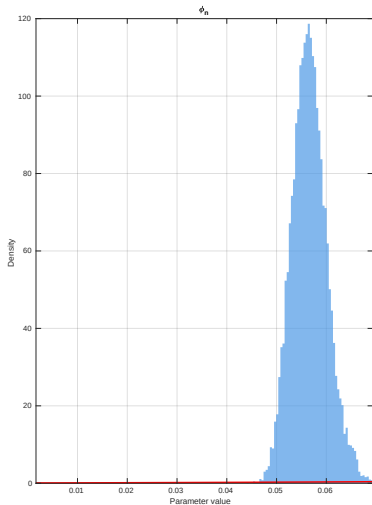
Other Shocks: Kim et al. (2025), Piffer and Podstawski (2018), Chahrour and Jurado (2022), Adams and Barrett (2024)

Posterior Distributions: Not Very Ricardian!



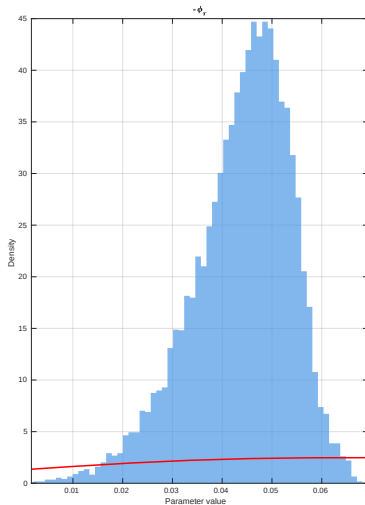
- θ : Uniform prior on model-based interval, $\mathbb{E}[\theta] = 0.19$

Posterior Distributions: Not Very Ricardian!



- θ : Uniform prior on model-based interval, $\mathbb{E}[\theta] = \mathbf{0.19}$
- MPC (coefficient on net worth n_{t-1}): $\mathbb{E}[1 - \beta] = \mathbf{0.056}$

Posterior Distributions: Not Very Ricardian!



- θ : Uniform prior on model-based interval, $\mathbb{E}[\theta] = \mathbf{0.19}$
- MPC (coefficient on net worth n_{t-1}): $\mathbb{E}[1 - \beta] = \mathbf{0.056}$
- MPC-DFT (coefficient on $-\tau_t$): $\mathbb{E}[(1 - \beta)(1 - \theta)] = \mathbf{0.045}$

Point Estimates: Not Very Ricardian!

Specification	Attenuation (θ)	MPC (ϕ_n)	MPC-DFT ($-\phi_\tau$)
Baseline	0.194 [0.019, 0.544]	0.056 [0.051, 0.063]	0.045 [0.025, 0.058]
Non-durable consumption	0.130 [0.013, 0.372]	0.058 [0.052, 0.064]	0.050 [0.036, 0.060]
Survey-based forecasts	0.231 [0.020, 0.657]	0.009 [0.007, 0.011]	0.007 [0.003, 0.010]
Non-separable utility	0.096 [0.029, 0.234]	0.065 [0.050, 0.072]	0.059 [0.039, 0.067]
Including SVAR instruments	0.123 [0.013, 0.262]	0.063 [0.059, 0.067]	0.055 [0.046, 0.062]
Six selected instruments	0.174 [0.017, 0.404]	0.033 [0.028, 0.037]	0.027 [0.019, 0.034]

Notes: Point estimates are medians of the marginal posterior distributions. The [5%, 95%] credible intervals are reported in brackets.

How Important is the Behavioral Friction Specifically?

- Our method nests other non-Ricardian mechanisms

How Important is the Behavioral Friction Specifically?

- Our method nests other non-Ricardian mechanisms
 - Isomorphic to many household-government discount wedges: OLG, occasionally-binding constraints, income taxes, simple hand-to-mouth consumers

How Important is the Behavioral Friction Specifically?

- Our method nests other non-Ricardian mechanisms
 - Isomorphic to many household-government discount wedges: OLG, occasionally-binding constraints, income taxes, simple hand-to-mouth consumers
 - By including additional controls: time-varying taxes, stochastic HtM income

How Important is the Behavioral Friction Specifically?

- Our method nests other non-Ricardian mechanisms
 - Isomorphic to many household-government discount wedges: OLG, occasionally-binding constraints, income taxes, simple hand-to-mouth consumers
 - By including additional controls: time-varying taxes, stochastic HtM income
- Requires additional structure to distinguish

How Important is the Behavioral Friction Specifically?

- Our method nests other non-Ricardian mechanisms
 - Isomorphic to many household-government discount wedges: OLG, occasionally-binding constraints, income taxes, simple hand-to-mouth consumers
 - By including additional controls: time-varying taxes, stochastic HtM income
- Requires additional structure to distinguish
 - Identify the household-government discount wedge from MPC

How Important is the Behavioral Friction Specifically?

- Our method nests other non-Ricardian mechanisms
 - Isomorphic to many household-government discount wedges: OLG, occasionally-binding constraints, income taxes, simple hand-to-mouth consumers
 - By including additional controls: time-varying taxes, stochastic HtM income
- Requires additional structure to distinguish
 - Identify the household-government discount wedge from MPC
 - Separately identify θ from MPC-DFT

How Important is the Behavioral Friction Specifically?

- Our method nests other non-Ricardian mechanisms
 - Isomorphic to many household-government discount wedges: OLG, occasionally-binding constraints, income taxes, simple hand-to-mouth consumers
 - By including additional controls: time-varying taxes, stochastic HtM income
- Requires additional structure to distinguish
 - Identify the household-government discount wedge from MPC
 - Separately identify θ from MPC-DFT
 - $\Rightarrow$ Behavioral friction causes $\sim 50\%$ non-Ricardianism

General Equilibrium

GE Model Summary

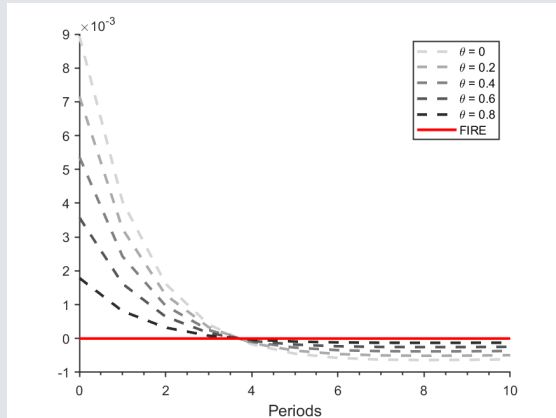
- Simple behavioral RBC:
 - Capital investment, inelastic labor, risk-free debt
 - Many shocks
 - Cognitive discounting over taxes *only*

GE Model Summary

- Simple behavioral RBC:
 - Capital investment, inelastic labor, risk-free debt
 - Many shocks
 - Cognitive discounting over taxes *only*
- Tax *cut* makes consumers think they are *richer*

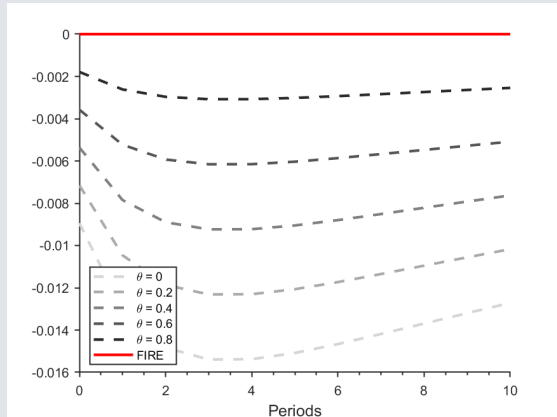
GE Model Summary

- Simple behavioral RBC:
 - Capital investment, inelastic labor, risk-free debt
 - Many shocks
 - Cognitive discounting over taxes *only*
- Tax *cut* makes consumers think they are *richer*
- $\implies$ consumption $\uparrow$

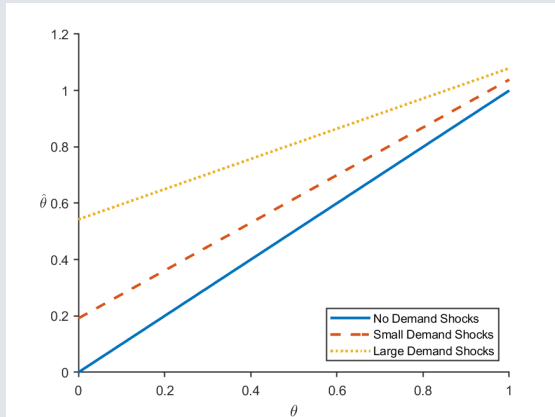


GE Model Summary

- Simple behavioral RBC:
 - Capital investment, inelastic labor, risk-free debt
 - Many shocks
 - Cognitive discounting over taxes *only*
- Tax *cut* makes consumers think they are *richer*
- $\implies$ consumption $\uparrow$
- $\implies$ investment $\downarrow \implies$ recession

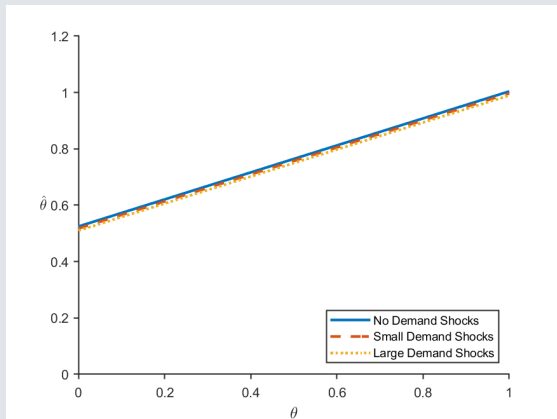


Estimates of the Model's Consumption Function



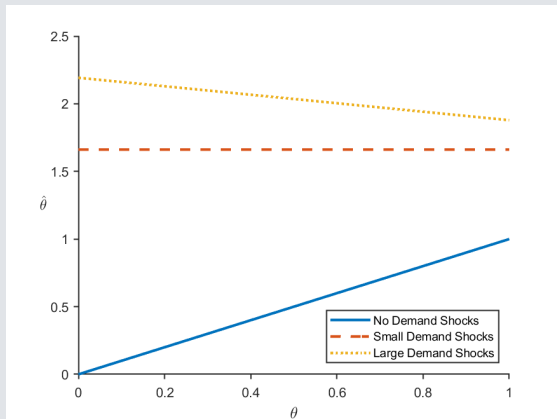
- OLS: only works without demand shocks

Estimates of the Model's Consumption Function



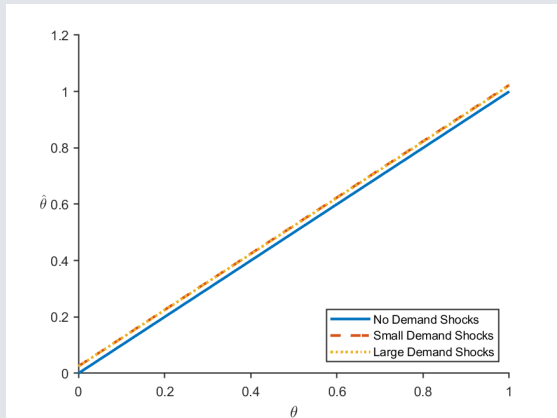
- OLS: only works without demand shocks
- Exog tax shocks: only works if Ricardian

Estimates of the Model's Consumption Function



- OLS: only works without demand shocks
- Exog tax shocks: only works if Ricardian
- Lagged aggregates: only works without demand shocks (or if i.i.d.)

Estimates of the Model's Consumption Function



- OLS: only works without demand shocks
- Exog tax shocks: only works if Ricardian
- Lagged aggregates: only works without demand shocks (or if i.i.d.)
- Macro IVs: always works!

Conclusion

- How Ricardian Are We? *Not very.*
- Behavioral non-Ricardianism is useful, likely, and doesn't require an otherwise non-rational model.
- Serious macro implications!
- **B-HIVE** extremely useful for “identifying modern macro equations with old shocks”
 - We'll release the code (easy to adapt!)
 - Please reuse the library of structural shocks (on my website)

References

Adams, Jonathan J. and Philip Barrett, “Shocks to Inflation Expectations,” *Review of Economic Dynamics*, October 2024, 54, 101234.

Aruoba, S. Borağan and Thomas Drechsel, “Identifying Monetary Policy Shocks: A Natural Language Approach,” May 2024.

Bauer, Michael D. and Eric T. Swanson, “A Reassessment of Monetary Policy Surprises and High-Frequency Identification,” *NBER Macroeconomics Annual*, May 2023, 37, 87–155.

Baumeister, Christiane and James D. Hamilton, “Structural Interpretation of Vector Autoregressions with Incomplete Identification: Revisiting the Role of Oil Supply and Demand Shocks,” *American Economic Review*, May 2019, 109